Gonchala Lakshmi Padmaja

BUSINESS SALES ANALYTICS REPORT

A data-driven review of sales, profitability, products, customers, marketing, delivery, returns and customer satisfaction.

Analysis scope: Financial performance sections use USD Completed and Paid orders unless otherwise stated. Operational delivery and returns sections use the full order dataset.

1. Executive Summary

The notebook analyzes an e-commerce sales dataset and builds an interactive Plotly dashboard. The validated USD financial subset contains **61,099 orders** from **14,680 customers**, generating **$74.97 million revenue** and **$29.56 million profit**. The resulting profit margin is **39.43%** and average order value is approximately **$1,227**.

Key Performance Indicator	Validated Result
Total Revenue (USD)	$74,968,389.31
Total Profit (USD)	$29,560,380.55
Completed & Paid USD Orders	61,099
Unique USD Customers	14,680
Quantity Sold	348,703
Average Order Value	$1,227.00
Profit Margin	39.43%
Average Historical Customer Value	$5,106.84

Headline findings: Electronics is the leading revenue category; the South is the largest region by USD revenue; Organic Search is the largest recorded marketing channel by revenue; and repeat customers account for 98.39% of USD revenue.

2. Dataset and Analysis Approach

The notebook combines the order-level sales data with customer, product catalogue and order-item data. Analysis is organized around sales and revenue, profitability, product categories, regions, customer segmentation, marketing, delivery, returns, ratings, loyalty and monthly cohorts.

The financial summary deliberately uses USD orders only. The source dataset contains multiple currencies, so combining raw amounts across currencies would not produce a meaningful consolidated financial total. Delivery and returns metrics are reported from the full order dataset and are not restricted to the USD subset.

The final financial validation in the notebook compares expected and actual revenue, profit, orders, customers and quantity. All five differences are zero (subject to floating-point display rounding), and the notebook prints **ALL FINAL FINANCIAL VALIDATIONS PASSED**.

3. Product and Category Performance

Product-level financial reconciliation was performed before the final category analysis. The notebook identifies additional product-item rows for some orders and constructs a reconciled product dataset to align product totals with validated order-level financials. The reconciliation is a data-preparation rule based on sequential product rows; excluded rows should be treated as suspected extra or inconsistent records rather than independently proven invalid transactions.

Category	Revenue (USD)	Profit (USD)	Orders	Margin
Electronics	$15,977,778.52	$4,830,373.96	9,942	30.23%
Jewelry	$9,819,156.42	$3,609,489.39	8,811	36.76%
Home Appliances	$9,785,924.97	$3,697,244.33	10,020	37.78%
Automotive	$6,408,999.00	$2,291,205.65	10,362	35.75%
Sports & Outdoors	$6,158,012.85	$2,617,166.62	11,499	42.50%

Electronics leads the listed categories in revenue and profit. Sports & Outdoors shows a higher profit margin among the five displayed categories. The notebook's category margin analysis also identifies Grocery as the highest-margin category at 52.44%.

4. Regional Performance

Region	Orders	Revenue (USD)	Profit (USD)	Margin
South	23,707	$29,129,528.27	$11,499,709.39	39.48%
Central	17,986	$21,964,526.92	$8,659,039.46	39.42%
East	13,091	$15,995,615.40	$6,302,621.39	39.40%
West	6,315	$7,878,718.72	$3,099,010.31	39.33%

The South region contributes the largest revenue and profit totals. Profit margins are tightly clustered across the four regions, ranging from 39.33% to 39.48%.

5. Customer Segmentation and Historical Value

The USD customer analysis includes 14,680 customers, averaging 4.16 orders per customer. The frequency-based segmentation records 1,794 Loyal customers, 6,955 Repeat customers, 4,920 Occasional customers and 1,011 One-Time customers.

RFM segmentation uses recency, frequency and monetary scores. The notebook assigns Champions (score 13 or higher), Loyal Customers (10–12), Potential Loyalists (7–9), Needs Attention (5–6) and At Risk (3–4). The segment labels are analytical groupings, not validated predictions of future churn.

RFM Segment	Customers	Revenue (USD)	Profit (USD)	Profit Share
Loyal Customers	4,108	$26,366,966.83	$10,412,114.16	35.22%
Champions	2,698	$24,689,262.50	$9,564,619.12	32.36%
Potential Loyalists	3,988	$15,772,766.92	$6,277,193.01	21.24%
Needs Attention	2,126	$5,597,179.63	$2,258,966.55	7.64%
At Risk	1,760	$2,542,213.43	$1,047,487.71	3.54%

Average historical customer value is $5,106.84. This is an observed-spend measure calculated from historical orders, not a predictive customer lifetime value model.

6. Loyalty and Cohort Retention

The notebook identifies 13,669 repeat customers and 1,011 one-time customers. The repeat purchase rate is 93.11%. Repeat customers generated $73,761,557.86 in revenue (98.39% of USD revenue), while one-time customers generated $1,206,831.45 (1.61%).

Cohort Month Index	Eligible Customers	Retained Customers	Retention Rate
1	14,638	974	6.65%
2	14,613	937	6.41%
3	14,595	873	5.98%
6	14,532	940	6.47%
12	14,368	988	6.88%

The cohort analysis uses the first observed USD order month as each customer's cohort and tracks activity in later calendar months. The retention percentages above are the notebook's reported month-index summary and should not be confused with the 93.11% repeat-purchase rate, which is a customer-level measure over the full observed period. Cohort order and customer counts reconcile to the USD dataset, and month-zero retention validates at 100%.

7. Marketing and Discount Analysis

Organic Search is the largest recorded marketing channel by USD revenue, with $14,966,966.30 revenue and $5,905,948.31 profit (19.96% revenue share). Google Ads contributes $11,300,423.40 revenue; Direct contributes $11,063,909.06. Channel profit margins are close, from 39.23% (Google Ads) to 39.70% (TikTok).

Marketing Channel	Orders	Revenue (USD)	Profit (USD)	Margin
Organic Search	12,282	$14,966,966.30	$5,905,948.31	39.46%
Google Ads	9,112	$11,300,423.40	$4,433,161.26	39.23%
Direct	9,046	$11,063,909.06	$4,371,198.41	39.51%
Facebook Ads	7,410	$9,115,504.83	$3,599,067.02	39.48%
Instagram	6,164	$7,489,560.38	$2,962,204.12	39.55%

Campaign names are missing for 36,694 of 61,099 USD orders (60.06%), representing $45,160,632.30 or 60.24% of revenue. Default_Campaign accounts for 10,249 orders and $12,445,872.82 revenue. Among specifically named campaigns, Google_Search_Q1 is the largest shown, with $1,563,725.40 revenue.

Total USD discounts are $16,693,346.08. Nearly all USD orders have a positive discount. The notebook reports lower observed profit margins in higher discount-dollar bands, but this is an association only: larger orders may also carry larger absolute discounts. Campaign/channel revenue does not establish return on ad spend because advertising spend data is not available.

8. Delivery and Shipping Performance

Operational delivery analysis covers 113,559 orders with recorded delivery outcomes. The on-time rate is 76.54%, delayed rate is 14.87%, and early rate is 8.59%. Average actual delivery time is 4.56 days versus an estimated 4.30 days.

Shipping Method	Orders	Avg Actual Days	Avg Estimated Days	On-Time Rate	Delay Rate
Same Day	11,460	0.82	0.49	76.81%	14.49%
Standard	50,924	5.25	5.00	76.63%	14.90%
Express	28,320	2.26	2.00	76.46%	14.96%

Shipping Method	Orders	Avg Actual Days	Avg Estimated Days	On-Time Rate	Delay Rate
Economy	22,855	7.74	7.50	76.30%	14.88%

Data-quality note: 24,557 orders have no delivery-day values; these are all Returned, Cancelled or Pending orders and are labelled Cancelled in delivery_status. Therefore delivery KPIs describe records with delivery outcomes, not every order in the dataset. Raw shipping-cost averages are not compared financially because the operational dataset contains multiple currencies.

9. Returns Analysis

There are 9,462 returned orders out of 138,116 total orders, giving an overall return rate of 6.85% using all orders as the denominator. All returned orders have a recorded return reason and there are no duplicate returned order IDs in the validation output.

Return Reason	Returned Orders	Share of Returns
Wrong Product	1,237	13.07%
Other	1,208	12.77%
Changed Mind	1,204	12.72%
Size Issue	1,182	12.49%
Defective Product	1,174	12.41%
Late Delivery	1,172	12.39%
Product Not as Expected	1,147	12.12%
Damaged Product	1,138	12.03%

Return rates by shipping method are close: Express 6.93%, Economy 6.88%, Standard 6.85% and Same Day 6.60%. These small differences are descriptive and do not establish that shipping method causes returns.

10. Customer Satisfaction

Customer ratings are available for 113,559 completed orders; 24,557 orders (17.78%) have no rating. The average rating is 3.68 out of 5 and the median is 3.70. The rating distribution classifies 67.91% as Satisfied and 22.15% as Highly Satisfied; 9.89% are Dissatisfied and 0.05% Very Dissatisfied.

Delivery Status	Rated Orders	Average Rating	Median Rating
Early	9,757	4.02	4.0
On Time	86,916	3.73	3.7
Delayed	16,886	3.22	3.2

The notebook reports a correlation of -0.3858 between delivery difference and customer rating. This indicates a negative association in the analyzed records, not proof of causation. Ratings are available only for completed orders in the dataset.

11. Interactive Dashboard

The notebook creates a four-row, two-column Plotly interactive business dashboard with KPI cards and sales-performance views. The export cell reports that the HTML dashboard was saved successfully and that the file-existence check returned True.

Dashboard file: Business_Sales_Interactive_Dashboard.html
Notebook-reported local location:
C:\Users\gowth\OneDrive\Documents\FUTURE_DS_01\dashboard\Business_Sales_Interactive_Dashboard.html

12. Recommendations and Next Steps

• **Protect and grow repeat-customer value:** Repeat customers contribute 98.39% of USD revenue. Use loyalty benefits and relevant retention campaigns while monitoring incremental results.

• **Investigate delivery delays:** Delayed deliveries have an average rating of 3.22 versus 3.73 for on-time and 4.02 for early deliveries. Review carrier, fulfilment and destination-level causes.

• **Improve campaign tracking:** Campaign names are absent for 60.06% of USD orders. Strengthen campaign tagging and capture advertising cost to support reliable ROAS analysis.

• **Review product accuracy:** Wrong Product is the most frequent recorded return reason. Audit product descriptions, images, variant selection and fulfilment checks.

• **Monitor category profitability:** Electronics leads revenue, but its 30.23% margin is below the overall 39.43% margin. Review product mix, sourcing cost, discounting and pricing.

• **Keep currency reporting separated:** Continue currency-specific reporting unless reliable exchange rates and a documented conversion date are applied.

• **Document product reconciliation:** Preserve original order-item data and record the sequential matching rule so the reconciled analysis remains reproducible and auditable.

13. Limitations

• The financial headline KPIs are USD-only and reflect Completed and Paid orders in the notebook's analysis subset.

• The report uses historical observed spending, not a predictive lifetime-value model.

• RFM segment labels and marketing-channel comparisons are descriptive, not validated causal or predictive models.

• Campaign tracking is incomplete, and ad-spend data is not available for ROAS or ROI calculation.

• Delivery and rating metrics exclude orders without completed delivery records; return rate uses all orders as its denominator.

• The product-item reconciliation rule aligns product totals with order totals but does not independently verify the business validity of every excluded row.

14. Conclusion

The analysis establishes a reconciled USD sales baseline, identifies major revenue and profit contributors, and highlights customer loyalty, delivery experience and marketing-data quality as important management areas. The notebook's final financial checks pass, and the interactive dashboard export is confirmed in the notebook output.

Source: Uploaded Business_Sales_Analysis Jupyter Notebook and its executed cell outputs. Figures in this report are transcribed from the notebook; interpretations and recommendations are descriptive and are qualified where the source data has limitations.